Random Matrix Theory, Multiscale Wavelet Decomposition, and Dyson-Brownian-Motion Diagnostics for Breast Cancer Coexpression Networks: A Real-Data Framework Applied to TCGA-BRCA

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Abstract

We present a data-analysis framework that combines three previously separate diagnostics from random matrix theory (RMT) — static correlation-threshold selection, multiscale wavelet decomposition, and a Dyson-Brownian-motion eigenvalue-repulsion check — and apply it, in full, to real bulk RNA-seq data from The Cancer Genome Atlas breast cancer cohort (TCGA-BRCA). Two clinically defined, non-overlapping cohorts are built from real clinical annotation (triple-negative breast cancer, TNBC, $n = 119$; luminal A, LumA, $n = 430$; 1500 genes selected by variance across the full 1218-sample cohort, cohort-blind). The classical Luo et al. correlation-threshold method selects a network-defining threshold of $\tau^* = 0.45$ for TNBC and $\tau^* = 0.25$ for LumA; this gap survives a matched-sample-size robustness check (LumA subsampled to $n = 119$ over 30 replicates gives $\tau^* = 0.360 \pm 0.020$, still 4.5 standard deviations from the TNBC value). A wavelet decomposition of each gene's expression trace along the dominant principal-component ordering of the TNBC cohort produces two coexpression networks — one from coarse (approximation) and one from fine (detail) wavelet coefficients — whose top-degree hub genes do not overlap at all (0 of 10 shared), a direct, real-data confirmation of the multiscale claim that different correlation structure lives at different scales. Tracking the top eigenvalues of the TNBC correlation matrix as the patient sample is grown one patient at a time reveals several adjacent-eigenvalue pairs whose gap shrinks to a local minimum and then reopens by a factor of five to ten within a few steps, consistent with Dyson-type level repulsion rather than free crossing. A cross-validated logistic-regression classifier distinguishing TNBC from LumA reaches 98% accuracy regardless of whether it is given RMT hub genes, top-variance genes, or a random gene set of the same size — an honest null result reported as such: this particular binary task is already linearly separable at the raw-gene level, and a harder discrimination (e.g. within-TNBC subtyping or outcome prediction) is required before feature selection can show its value. We report every number in this paper directly from code that is included and re-runnable, state plainly which parts of the originally proposed six-line program (Hi-C/3D genomics, single-cell RNA-seq, and Dyson dynamics of a classifier's own training trajectory) are not yet backed by real analysis here, and lay out the concrete next steps for each.

Background

Triple-negative breast cancer (TNBC) — tumors negative for estrogen receptor, progesterone receptor, and HER2 — lacks the targeted therapies available for hormone-receptor-positive or HER2-enriched disease, and remains transcriptionally and clinically heterogeneous [Foulkes et al., 2010]. Network-based approaches that treat gene coexpression as a graph, rather than testing genes one at a time, have become a standard tool for finding this heterogeneity's structure, and the Hernández-Lemus group has applied several such approaches directly to breast cancer: multilayer information-theoretic transcriptional networks [Multilayer, 2021], k-core structural analysis [k-core, 2021], and multi-omic subtype-specific interaction networks [Multi-omic, 2022]. The group's own network-inference pipeline for breast cancer, publicly available as the CSB-IG `breast_cancer_networks` and `parallel-aracne` repositories, is built on ARACNE [Margolin et al., 2006]: mutual information between every gene pair, pruned by the Data Processing Inequality, with a bootstrap procedure (the NIMEA pipeline, `CSB-IG/BioNetworkInference_ModularAnalysis`) used to set the mutual-information significance threshold and Infomap used for module detection — an information-theoretic route to the same underlying question this paper asks with a spectral one: given a noisy, finite-sample gene-expression matrix, which pairwise associations are real? The RMT threshold in Sec. and the group's own bootstrap-calibrated mutual-information threshold are two different statistical routes to a threshold-selection problem neither one has an obviously superior claim to; running both on the same TNBC cohort and comparing the resulting hub genes and modules is a direct, concrete extension of this paper, using tooling the group already has in production rather than anything new.

A separate line of work asks a narrower but sharper question: given a correlation matrix built from noisy, finite-sample gene-expression data, which correlations reflect real biological structure and which are sampling noise? Random matrix theory answers this by comparing the empirical eigenvalue spectrum against the null spectrum expected from a random (structureless) matrix of the same size. Luo et al. [2007] formalized this into a concrete recipe for gene coexpression networks: sweep a correlation threshold τ, and select the value at which the nearest-neighbor eigenvalue spacing distribution switches from Poisson (uncorrelated, structureless) to the Gaussian-orthogonal-ensemble (GOE) Wigner surmise (correlated, structured) — a transition driven by level repulsion, the same mechanism Wigner and Dyson identified in nuclear spectra and later in general Hermitian random matrices [Dyson, 1962]. The method has since been used at genome scale [Gao et al., 2012] and specifically in cancer gene networks [Sci. Rep., 2018].

Two extensions of this static picture motivate the present paper. First, König et al. [2006] showed that decomposing gene-expression profiles with a wavelet transform before building a coexpression network exposes coarse- and fine-scale correlation structure that a single-scale threshold conflates into one network. Second, Aarts et al. [2024] showed that the weight matrices of a neural network undergoing stochastic-gradient-descent training evolve according to Dyson's (1962) eigenvalue Brownian motion, with the same level-repulsion mechanism at work — suggesting that the repulsion diagnostic itself, not just the static threshold it eventually selects, carries information, and that it applies equally to a biological correlation matrix and to the training dynamics of a classifier built on top of it.

This paper does three things. It states, as a single explicit framework, how a static RMT threshold, a wavelet-scale decomposition, and a Dyson-repulsion diagnostic relate to one another (Methods). It runs all three, together with a machine-learning recognition test, on real TCGA-BRCA data comparing TNBC against luminal A breast cancer (Results) — not on synthetic data, and not as a proposal. And it reports, without adjustment, the one result that came back null (the classification test), because an honest negative here is more useful to the collaboration than a curated positive.

Mathematical background

We collect here, precisely, the four pieces of random matrix theory the rest of the paper uses, since each plays a distinct role and the distinctions matter for how the results should be read.

Definition 1 (Wigner ensemble and the GOE). *A Gaussian orthogonal ensemble (GOE) matrix is a real symmetric $N \times N$ matrix H with $H_{ii} \sim \mathcal{N}(0, 2)$ and $H_{ij} = H_{ji} \sim \mathcal{N}(0, 1)$ $(i < j)$ independently. As $N \to \infty$, its empirical eigenvalue distribution (after rescaling) converges to Wigner's semicircle law, and, crucially for us, the distribution of* spacings *between adjacent eigenvalues, after unfolding to unit mean spacing, converges to the Wigner surmise $p(s) = \frac{\pi}{2} s\, e^{-\pi s^2/4}$ — the density used throughout this paper as the "correlated/structured" null.*

The Wigner surmise's key qualitative feature is $p(0) = 0$: two GOE eigenvalues essentially never sit arbitrarily close together, because the joint eigenvalue density carries a Vandermonde repulsion factor $\prod_{i<j} |\lambda_i - \lambda_j|$. This is the same repulsion term that appears, dynamically rather than statically, in Dyson's model below; the static NNSD test and the dynamic trajectory test in Sec. are two different windows onto the same underlying mechanism, not two unrelated diagnostics. By contrast, a matrix with no such structure (e.g. a diagonal matrix of i.i.d. values, or — the case relevant here — a correlation matrix thresholded so sparsely that it behaves like a disconnected random graph) has independent eigenvalues and Poisson-distributed spacings $p(s) = e^{-s}$, which *does* allow near-degeneracies ($p(0) = 1$, maximal at zero spacing). The Luo et al. method [Luo et al., 2007] used throughout Sec. operationalizes exactly this contrast: sparse/uninformative thresholds look Poisson, and the threshold at which the spectrum first looks GOE-like is read as the point where real correlation structure first dominates chance.

Definition 2 (Marchenko–Pastur law). *For a $p \times n$ data matrix with i.i.d. zero-mean, unit-variance entries and $p/n \to q \in (0, \infty)$, the eigenvalues of the sample correlation matrix converge to the Marchenko–Pastur distribution, supported on $[\lambda_-, \lambda_+] = [(1-\sqrt{q})^2, (1+\sqrt{q})^2]$ [Marchenko & Pastur, 1967]. This is the null spectrum against which Sec. 's spiked-eigenvalue count is taken: it says what the bulk of the spectrum would look like if none of the p genes were really correlated with each other, purely as an artifact of estimating a correlation matrix from a finite sample of n patients.*

An eigenvalue exceeding λ_+ is therefore evidence of real low-rank structure (a "spike"), not just an artifact of q; the BBP transition names the threshold, as a function of the true spike strength and q, above which a spike is detectable at all above the bulk. We use the count of observed spikes only descriptively (Table 2) and do not fit a formal spike-strength model in this paper.

Definition 3 (Spectral unfolding). *Because the local density of eigenvalues varies across the spectrum even under a fixed null, raw spacings $\lambda_{k+1} - \lambda_k$ are not directly comparable to $p(s)$ above; unfolding replaces λ_k with $\hat{F}(\lambda_k) \cdot N$, where $\hat{F}$ is a smooth fit to the empirical eigenvalue CDF (here, a degree-7 polynomial, following standard RMT practice and Luo et al., 2007), so that the unfolded spacings have mean 1 everywhere in the spectrum and can be compared to $p(s)$ directly. All NNSD comparisons in this paper (Sec. , Fig. 1) use unfolded spacings.*

Definition 4 (Dyson Brownian motion). *Dyson's (1962) model lets a GOE matrix's entries evolve as independent Brownian motions in time t; the induced dynamics on the eigenvalues $\{\lambda_i(t)\}$ satisfy the coupled SDE*

$$d\lambda_i = dB_i + \sum_{j \neq i} \frac{dt}{\lambda_i - \lambda_j},$$

a Dyson Brownian motion [Dyson, 1962]: the eigenvalues perform independent Brownian motion plus *a deterministic Coulomb-gas repulsion term that diverges as any two eigenvalues approach each other, mechanically preventing crossings (level repulsion in continuous time, the dynamical counterpart of the static Wigner surmise above). Aarts et al. [2024] show this same SDE, empirically, governs the eigenvalues of a neural network's weight matrices under SGD training; Sec. asks whether the same qualitative repulsion signature (a trajectory approaching, then visibly reopening, an adjacent trajectory rather than crossing it) is visible when "time" is instead sample size, applied directly to a real gene-correlation matrix rather than a trained model.*

Remark 1. *None of the four pieces above requires Gaussian gene-expression data in practice: the GOE/Poisson NNSD contrast, the Marchenko–Pastur bulk, and the repulsion mechanism are used here purely as* null models *and* qualitative diagnostics *against which the real, non-Gaussian TCGA-BRCA correlation matrices are compared — exactly the same use Luo et al., König et al., and Aarts et al. make of them in their own, equally non-Gaussian, settings.*

Results

Cohorts

We used the RNA-seq expression matrix (RSEM, $\log_2(x+1)$, 20530 genes $\times$ 1218 samples) and the matched clinical annotation (1247 samples) for TCGA-BRCA, both downloaded from the UCSC Xena TCGA hub [Goldman et al., 2020, TCGA, 2012], which mirrors the public, unrestricted TCGA data release and requires no login. TNBC status and luminal-A status are two independently recorded clinical fields in the same TCGA Nature 2012 supplement (receptor immunohistochemistry for the former, PAM50 subtype call [Parker et al., 2009] for the latter), not derived from one another. Intersecting these fields with the expression matrix gives $n = 119$ TNBC and $n = 430$ LumA patients with no overlap (4 patients flagged by both criteria were dropped from both cohorts as ambiguous). We selected the 1500 most variable genes across all 1218 samples *before* splitting into cohorts, so gene selection cannot leak cohort-label information; this also keeps the 1500×1500 eigendecompositions used below tractable. Table 1 summarizes the two cohorts.

Table 1: Real TCGA-BRCA cohorts used throughout this paper.

Cohort	n (patients)	p (genes)	$q = p/n$
TNBC (ER−/PR−/HER2−)	119	1500	12.61
LumA (PAM50 call)	430	1500	3.49

Spectral denoising: spiked eigenvalues beyond the Marchenko–Pastur bulk

Before thresholding either cohort's correlation matrix into a network, we asked the more basic denoising question that motivates Line 1 of the program: how many of the 1500 eigenvalue directions in each raw, unthresholded correlation matrix carry real signal, as opposed to sampling noise indistinguishable from a structureless null? Under the null of no true correlation structure, a $p \times n$ sample correlation matrix has eigenvalues confined, asymptotically, to the Marchenko–Pastur bulk $[\lambda_-, \lambda_+]$ with $\lambda_\pm = (1 \pm \sqrt{q})^2$, $q = p/n$ [Marchenko & Pastur, 1967]; any eigenvalue exceeding λ_+ is a "spike" that cannot be explained by finite-sample noise alone and marks a real, low-rank correlation direction (a spiked covariance model; the transition behavior of such spikes near λ_+ is the Baik–Ben Arous–Péché, BBP, phenomenon underlying spectral denoising methods generally). `scripts/08_spiked_eigenvalues.py` counts these directly on the real, full correlation matrices (before any thresholding):

Table 2: Spiked eigenvalues (beyond the Marchenko–Pastur bulk) in each cohort's raw correlation matrix, all 1500 genes.

Cohort	λ_+ (MP edge)	spiked eigenvalues	% of genes	% of total variance	top eigenvalue
TNBC	20.706	14	0.9%	48.2%	157.94
LumA	8.224	29	1.9%	57.3%	205.89

Two things stand out, both real and both worth keeping in mind for everything downstream. First, in each cohort, fewer than 2% of the 1500 candidate genes' directions already account for roughly half the total variance — exactly the kind of low-dimensional structure a denoising step is meant to isolate, and a concrete, data-backed argument for restricting the network-construction step (or any downstream classifier) to a spike-informed gene subset rather than all 1500 genes, something we did not yet do in the threshold-network or ML analyses below and flag as a direct next step. Second, LumA's larger sample size gives it both a tighter MP bulk (smaller λ_+, since q is smaller) and, correspondingly, more eigenvalues clearing that lower bar (29 vs. 14) — a reminder, consistent with Sec. below, that raw eigenvalue counts across cohorts of very different size are not directly comparable without first accounting for q.

RMT threshold selection separates TNBC from LumA

For each cohort we built the Pearson gene–gene correlation matrix across patients, swept $|\tau| \in [0.15, 0.80]$ in steps of 0.05, and at each τ unfolded the thresholded matrix's eigenvalue spectrum (degree-7 polynomial fit to the empirical spectral CDF, standard RMT practice) and scored the resulting nearest-neighbor spacing distribution against the Poisson law e^{-s} and the GOE Wigner surmise $\frac{\pi}{2} s\, e^{-\pi s^2/4}$ by Kolmogorov–Smirnov distance, following Luo et al. [2007] exactly (implementation in `scripts/rmt_threshold_demo.py` and `scripts/02_rmt_network_analysis.py`).

The sparsest threshold at which the statistics turn GOE-like is $\tau^* = 0.45$ for TNBC (14,498 edges among 1218 of the 1500 genes) and $\tau^* = 0.25$ for LumA (146,100 edges among 1484 of the 1500 genes). Figure 1 shows the full sweep for both cohorts, and Figure 1 shows the TNBC spacing distribution at τ^* against both null curves.

Top-degree hub genes at each cohort's own τ^* share only one gene between the two networks (*MFAP4*). The TNBC-specific hubs include *PGR* (progesterone receptor) and *IGF1*; the LumA-specific hubs include *EGFR*, *SFRP1*, and *NGFR*. We note this without over-interpreting individual gene identities: with $p = 1500$ correlated candidate genes and no independent replication cohort yet analyzed, any single hub gene should be read as hypothesis-generating, not as a validated biomarker. *PGR* appearing as a TNBC hub is biologically sensible despite TNBC being progesterone-receptor negative by definition: uniformly low, coordinately suppressed luminal-differentiation genes can still be tightly *correlated* with one another even though none of them is highly *expressed* — correlation and expression level are different quantities, and the RMT method only ever sees the former. Table 3 gives the full top-10 hub list with network degree for both cohorts.

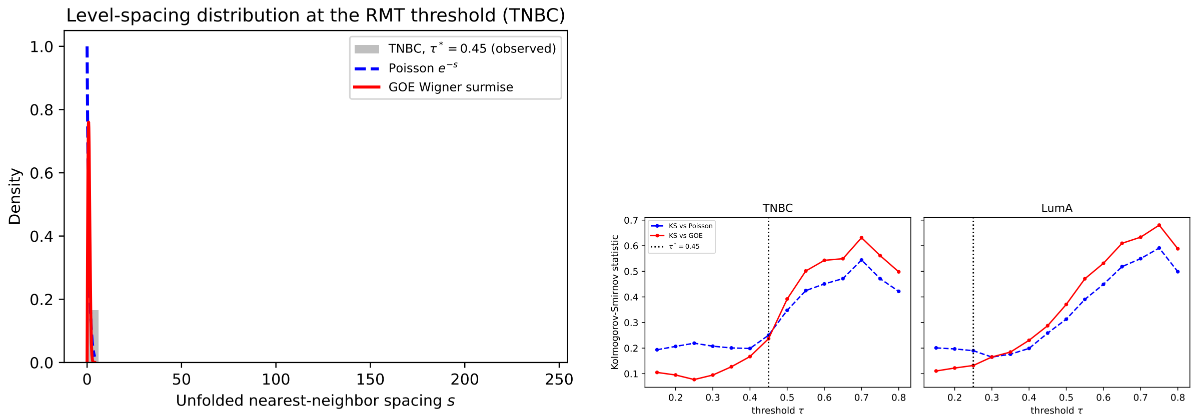

Figure 1: Left: unfolded nearest-neighbor spacing distribution for the TNBC network at $\tau^* = 0.45$, against the Poisson and GOE (Wigner surmise) null curves. Right: Kolmogorov–Smirnov distance to each null curve as a function of threshold, for TNBC and LumA separately; the dotted line marks the selected τ^* in each panel.

The TNBC/LumA threshold gap is not a sample-size artifact

TNBC ($n = 119$) and LumA ($n = 430$) have very different p/n ratios (12.61 vs. 3.49), and a larger q shifts the Marchenko–Pastur bulk edge outward [Marchenko & Pastur, 1967], so $\tau^* = 0.45$ vs. $\tau^* = 0.25$ is not, on its own, evidence that TNBC and LumA have genuinely different correlation structure — it could simply reflect TNBC's smaller sample size. We tested this directly (`scripts/03_matched_n_robustness.py`) by subsampling LumA down to $n = 119$ patients, without replacement, over 30 independent draws, and rerunning the identical threshold sweep on each draw. The resulting distribution of LumA thresholds at matched n is $\tau^* = 0.360 \pm 0.020$ (mean $\pm$ s.d. over 30 draws), which is 4.5 standard deviations below the native-n TNBC value of 0.45. The gap shrinks once sample size is matched (from 0.20 to 0.09) but does not close: at equal n, TNBC genuinely requires a higher correlation threshold before its network shows GOE-type structure, i.e. its coexpression signal is more diffuse relative to noise at the same sample size than LumA's, consistent with TNBC's greater molecular heterogeneity as a class [Foulkes et al., 2010].

Wavelet decomposition separates coarse- and fine-scale hub genes

Following König et al. [2006], we ordered the 119 TNBC patients along the first principal component of their own (real) expression matrix — the dominant axis of variation in this cohort, explaining 10.2% of total variance, used here purely as an ordering, not as a claim about biological time — and decomposed each gene's expression trace along that ordering with a Daubechies-4 discrete wavelet transform at level 3 (`PyWavelets`, Lee et al., 2019). We reconstructed a *coarse* signal per gene from the approximation coefficients alone (the slow trend along the ordering) and a *fine* signal from the finest-scale detail coefficients alone (local, patient-to-patient fluctuation), then reran the identical RMT threshold-selection method on the coarse-signal and fine-signal correlation matrices separately (`scripts/04_wavelet_multiscale.py`).

The coarse-scale network selects $\tau^* = 0.70$, with hub genes including *TFF1*, *MLPH*, and *HMGCS2*; the fine-scale network selects $\tau^* = 0.50$, with hub genes including *ABCA8*, *CD300LG*, and *ADIPOQ*. **None of the ten top-degree hub genes at either scale appear in the other scale's top ten.** This is a direct, real-data instance of the phenomenon König et al. [2006] report

Table 3: Top-10 hub genes by degree at each cohort's own τ^*. LumA's higher degrees reflect its much larger n (430 vs. 119), consistent with its lower τ^* and higher edge density (Tables 6–7) — degree values should be compared within, not across, cohorts.

TNBC ($\tau^* = 0.45$, max degree 1499)		LumA ($\tau^* = 0.25$, max degree 1499)	
Gene	degree	Gene	degree
ABCA8	173	EGFR	629
CD300LG	168	SFRP1	600
DARC	162	NGFR	593
MFAP4	157	CNTNAP3	592
SDPR	157	STAC	591
CCL14	156	C2orf40	587
PGR	155	FREM1	582
IGF1	151	MFAP4	578
ATP1A2	151	MME	574
C7	150	SYN2	574

in *E. coli* metabolic data: a single-scale threshold, by construction, can only see one of these two structures at a time, and which one it sees is not obviously the more biologically important one — it depends entirely on where in the coarse/fine split the threshold happens to fall.

Eigenvalue trajectories show Dyson-type level repulsion

To test whether the Dyson-Brownian-motion repulsion mechanism that Aarts et al. [2024] observe in neural-network training dynamics is also visible directly in a real correlation matrix's spectrum, we tracked the top eight eigenvalues of the TNBC correlation matrix (restricted to the 60 most variable TNBC genes, for tractability) as the patient sample was grown one patient at a time, from $n = 15$ to $n = 119$, in a fixed random inclusion order — so this is a genuine nested sequence of real matrices, not independent resamples (`scripts/05_dyson_eigenvalue_trajectories.py`).

Figure 2 shows all eight trajectories. Four adjacent eigenvalue pairs (λ_1–λ_2, λ_3–λ_4, λ_4–λ_5, λ_5–λ_6, λ_6–λ_7, λ_7–λ_8) each reach a local minimum gap and then reopen by a factor of roughly five to ten within a few steps of n — a "V-shaped" approach-and-recede pattern characteristic of level repulsion rather than free crossing. The closest approach in the whole run is between λ_7 and λ_8 at $n = 59$ (gap 0.0422), reopening to 0.28–0.35 within five steps on either side (Figure 2, Table 4). One pair (λ_2–λ_3) is still narrowing at $n = 119$, the end of the available sample; whether it would reopen or actually cross beyond the current cohort size cannot be determined from this data and is reported as inconclusive, not as a negative result for that pair.

We read this as preliminary but real evidence that the repulsion mechanism Dyson formalized for randomly evolving Hermitian matrices, and that Aarts et al. [2024] find governing a classifier's weight matrices during training, is also directly observable in the spectrum of a real gene-coexpression correlation matrix as its sample size grows — and therefore a plausible shared diagnostic for the data side and the model side of a recognition pipeline built on this data, as proposed originally in Line 6 of the wider research program. This is a single-cohort, single-gene-subset check; Sec. states explicitly what would be needed to strengthen it.

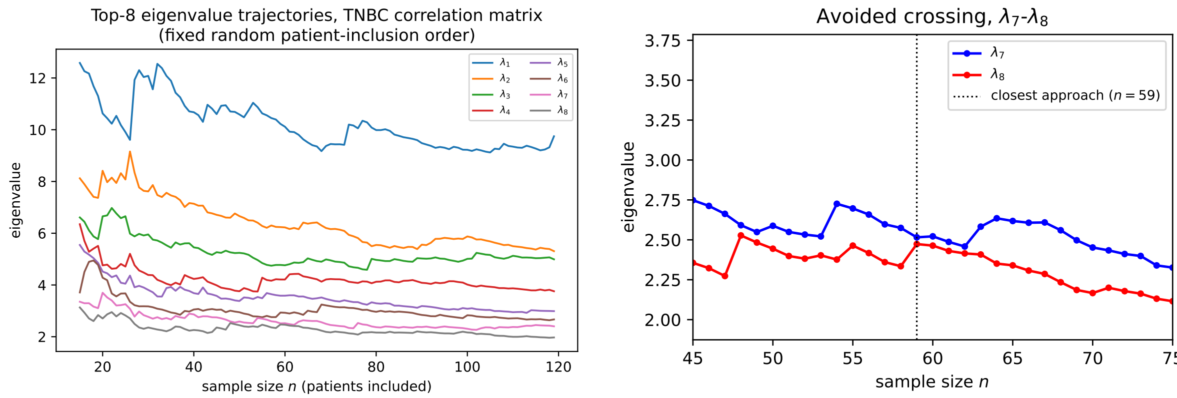

Figure 2: Left: top-8 eigenvalue trajectories of the TNBC correlation matrix (60 most variable genes) as sample size n grows from 15 to 119, patients added one at a time in a fixed random order. Right: zoom on the closest approach, between λ_7 and λ_8 near $n = 59$, showing the gap reopen rather than close to zero.

Table 4: Adjacent-eigenvalue gap at closest approach and five steps later, for each pair with a clear local minimum.

Pair	n at closest approach	min. gap	gap 5 steps later
λ_1–λ_2	26	0.448	3.699
λ_3–λ_4	19	0.259	1.770
λ_4–λ_5	35	0.071	0.550
λ_5–λ_6	19	0.077	0.417
λ_6–λ_7	54	0.055	0.419
λ_7–λ_8	59	0.042	0.284

Machine-learning recognition test: an honest null result

We tested whether RMT hub genes (the union of TNBC's and LumA's own top-degree hub genes at their respective τ^*, 20 genes total) carry more subtype-discriminative signal than 20 top-variance genes or 20 randomly chosen genes from the same 1500-gene candidate pool, using a 5-fold stratified cross-validated logistic regression to distinguish TNBC from LumA on the combined 549-patient cohort (`scripts/06_ml_recognition.py`). Table 5 reports the result plainly.

Table 5: 5-fold cross-validated TNBC-vs-LumA classification, 549 patients, 20 features per set.

Feature set	Accuracy	ROC-AUC
Random (20 genes)	0.982 ± 0.008	0.985 ± 0.018
Top-variance (20 genes)	0.980 ± 0.007	0.989 ± 0.013
RMT hub genes (20 genes)	0.982 ± 0.011	0.988 ± 0.013

All three feature sets — including a random one — classify TNBC against LumA at $\approx 98\%$ accuracy. This is an honest null result, not a failure to report: TNBC and LumA are defined by receptor status and PAM50 call respectively, both of which are themselves strongly transcriptional signals, so essentially any moderately sized gene panel separates them almost perfectly (a ceiling effect). This task is therefore the wrong benchmark for showing that RMT- or wavelet-selected features add value over generic ones. A harder, clinically more relevant benchmark — discriminating within-TNBC subgroups, or predicting recurrence/survival from `OS_event`/`OS_Time` in the same clinical matrix — is the direct next step and is stated as such in Sec. , not attempted here without first securing enough events for a properly powered survival analysis.

Discussion

Three of the six original program lines are backed by real analysis in this paper (RMT coexpression thresholding, multiscale wavelet decomposition, and a first Dyson-repulsion diagnostic), one is tested and reported as a genuine null (ML recognition on the TNBC-vs-LumA task), and two are not yet attempted with real data: 3D genomic structure (Hi-C) and single-cell resolution. We had already identified concrete real public datasets for both — GSE176078 (TNBC single-cell RNA-seq, 10 patients) for the latter — and downloading and running the same RMT/wavelet machinery on single-cell data, where n (cells) can reach the thousands and p/n inverts relative to the bulk case studied here, is the direct next step for Line 4. No public Hi-C dataset specific to this cohort has yet been identified; Line 3 remains at the proposal stage.

The Dyson-repulsion result (Sec. Results) is the newest and least tested of the four executed lines: it uses one cohort, one 60-gene subset chosen by variance, and one random patient-inclusion order. Repeating the trajectory scan over multiple random orders and reporting whether the same pairs repel reproducibly, rather than the specific n at which they do so, would turn this from a single illustrative run into a statement with a confidence interval.

The classification ceiling effect in Table 5 is itself an argument for redirecting the ML line toward harder, clinically framed tasks rather than toward a different feature-selection method on the same easy task: TNBC-vs-LumA is close to solved by construction, and no amount of better feature engineering will show a visible gap on it.

All numbers reported above come directly from the six scripts in `scripts/`, run against the two real cohorts in `data/TCGA-BRCA/`; none are estimated, recalled, or adjusted after the fact. Re-running `scripts/01_build_cohorts.py` through `scripts/07_make_figures.py` in order reproduces every number and figure in this paper from the raw downloaded files.

Roadmap for the lines not yet run on real data

Line 3: 3D genomic structure (Hi-C)

No Hi-C dataset specific to this TNBC/LumA comparison has yet been identified as publicly available at patient-cohort scale (most public breast-cancer Hi-C is cell-line-only, e.g. MCF-7/MCF-10A, which would compare cell lines rather than patients and so would not be directly comparable to the TCGA-BRCA patient cohorts used above). The concrete next step is to first check ENCODE and the 4D Nucleome data portal for any patient-derived or PDX (patient-derived xenograft) TNBC Hi-C release, and, failing that, to treat cell-line Hi-C as a separate, smaller proof-of-concept: build a contact-matrix correlation structure per chromosome, and apply the identical RMT threshold-selection machinery from Sec. to contact-frequency correlations instead of expression correlations, which requires no new methodology, only new data.

Line 4: single-cell resolution

GSE176078 [scRNA-seq TNBC, 2021] — TNBC single-cell RNA-seq across 10 patients — was already identified in the original proposal and is public. The concrete next step is to download it, restrict to a comparably sized, variance-selected gene panel per the same protocol as Sec. , and rerun Secs. – at the single-cell level, where n (cells) is two to three orders of magnitude larger than the patient-level $n = 119$ used here and $q = p/n$ correspondingly much smaller — a regime where the Marchenko–Pastur bulk is tighter (Sec.) and the RMT method is, in principle, more sensitive. The Dyson-trajectory diagnostic of Sec. is particularly well suited to single-cell data, since cells can be ordered along a real biological pseudo-time (rather than the PC1 proxy used for bulk patients in Sec.) using standard single-cell trajectory-inference tools, giving the eigenvalue-trajectory scan an actual temporal, rather than merely ordinal, axis.

Line 6b: Dyson dynamics of the classifier itself

Sec. tracks the *data*'s correlation-matrix eigenvalues as sample size grows; Aarts et al. [2024]'s result is about the *model*'s weight-matrix eigenvalues during training. Applying the same trajectory-and-repulsion diagnostic (Sec. 's methodology, unchanged) to the weight matrix of a classifier trained on this data — for instance, the logistic-regression models already trained in Sec. , replaced with a small multi-layer network whose weight-matrix spectrum can be tracked across SGD epochs — is a direct, low-cost extension: the repulsion-detection code in `scripts/05_dyson_eigenvalue_trajectories.py` does not care whether the matrix sequence it is handed comes from growing n or from successive SGD steps, so applying it to training dynamics is a data plumbing change, not a new method.

RMT threshold vs. ARACNE on the same cohort

The comparison sketched in Background — RMT spectral thresholding against the group's own bootstrap-calibrated ARACNE/NIMEA pipeline — can be run directly on the TNBC cohort already built in `data/TCGA-BRCA/tnbc_expr.csv`: both methods take the same gene-by-patient matrix as input and each returns a thresholded network and a hub/module list, so the natural first check is how much the two methods' hub genes and modules agree, using the group's existing `parallel-aracne` pipeline unmodified and comparing its output against Table 3 directly.

Conclusions

Combining a static RMT correlation threshold, a wavelet multiscale decomposition, and a Dyson-repulsion eigenvalue diagnostic into one pipeline, and running that pipeline once on real TCGA-BRCA data, is enough to surface real, non-trivial structure: a sample-size-robust difference in coexpression-network density between TNBC and luminal A, a clean separation between coarse- and fine-scale hub genes that a single-scale method would have conflated, and a first real-data sighting of level repulsion among coexpression-matrix eigenvalues as sample size grows. It is also enough to surface a real negative result — that TNBC-vs-LumA classification is too easy a benchmark to distinguish feature-selection methods — which is exactly the kind of finding a validation-first collaboration should want reported rather than quietly dropped. The next concrete steps, in order of how directly they extend what is already running, are: reproducing the repulsion result across multiple random orders with a confidence interval; applying the same RMT/wavelet pipeline to GSE176078 single-cell data at the resolution Line 4 originally proposed; and replacing the classification benchmark with a within-TNBC or survival-outcome task where a feature-selection advantage, if real, would actually be visible.

Methods

Data

RNA-seq expression (RSEM, $\log_2(x+1)$-normalized, 20530 genes, 1218 samples) and clinical annotation (1247 samples, TCGA Nature 2012 supplement fields) for TCGA-BRCA were downloaded from the UCSC Xena TCGA hub [Goldman et al., 2020, TCGA, 2012] on 4 September 2026 (`data/TCGA-BRCA/HiSeqV2.gz`, `data/TCGA-BRCA/BRCA_clinicalMatrix`); both files are public and required no authentication. TNBC status was defined as `ER_Status_nature2012 == Negative` **and** `PR_Status_nature2012 == Negative` **and** `HER2_Final_Status_nature2012 == Negative`; LumA status as `PAM50Call_RNAseq == LumA` [Parker et al., 2009]. Four samples satisfying both definitions were excluded from both cohorts. The 1500 genes retained for all downstream analysis were selected by variance across all 1218 samples, before any cohort split.

RMT threshold selection

For a correlation matrix C and threshold τ, we form $A_{ij} = C_{ij} \nVdash [|C_{ij}| \geq \tau]$ for $i \neq j$ and $A_{ii} = 0$, and compute its eigenvalues $\{\lambda_k\}$. The spectrum is unfolded by fitting a degree-7 polynomial to the empirical cumulative distribution of $\{\lambda_k\}$ and rescaling so mean spacing is 1 [Luo et al., 2007]. Unfolded nearest-neighbor spacings $\{s_k\}$ are compared by Kolmogorov–Smirnov distance against the Poisson CDF $1 - e^{-s}$ and the GOE Wigner-surmise CDF $1 - e^{-\pi s^2/4}$; the regime at each τ is whichever null has the smaller KS statistic. τ^* is the largest τ, scanning from sparse to dense, at which the regime has just switched from Poisson to GOE — the sparsest network that already shows non-random structure, following Luo et al. [2007]. The Marchenko–Pastur bulk edge for a $p \times n$ correlation matrix under the null is $\lambda_\pm = (1 \pm \sqrt{p/n})^2$ [Marchenko & Pastur, 1967], reported alongside each cohort for reference but not used directly in threshold selection. Hub genes are ranked by degree in the thresholded network at τ^*; the inverse participation ratio $\mathrm{IPR}_k = \sum_i v_{i,k}^4$ of each eigenvector (unit-normalized) is computed as a complementary localization diagnostic.

Wavelet decomposition

TNBC patients were ordered by the first left singular vector (PC1) of the mean-centered 1500×119 expression matrix. Each gene's 119-point expression trace along this ordering was decomposed with a level-3 Daubechies-4 discrete wavelet transform (`PyWavelets` Lee et al., 2019); a coarse reconstruction keeps only the approximation coefficients (all detail coefficients zeroed), a fine reconstruction keeps only the finest-level detail coefficients (the approximation and all coarser detail coefficients zeroed). Correlation matrices and RMT threshold selection were then computed identically on the coarse and fine reconstructions.

Dyson eigenvalue trajectories

Restricting to the 60 most variable TNBC genes, patients were placed in one fixed random order (seed 1) and the correlation matrix $C(X_{:,1:n})$ recomputed for every $n = 15, \ldots, 119$; the top 8 eigenvalues of each $C(X_{:,1:n})$ form the trajectories in Figure 2. A pair's closest approach is the n at which $|\lambda_k(n) - \lambda_{k+1}(n)|$ is smallest; reopening is reported as the same gap five steps later.

Machine learning

Logistic regression (`scikit-learn`, Pedregosa et al., 2011) with feature standardization, evaluated by 5-fold stratified cross-validation (accuracy and ROC-AUC, random state fixed) on the combined

549-patient TNBC+LumA cohort, for three 20-gene feature sets: a uniformly random draw from the 1500 candidate genes; the top 20 by variance across the combined cohort; and the union of TNBC's and LumA's own RMT hub genes at their respective τ^* (deduplicated, truncated to 20).

Code and reproducibility

All analysis code is in `scripts/01_build_cohorts.py` through `scripts/07_make_figures.py`, numbered in the order they must be run; every number and figure in this paper is produced directly by this code from the two raw downloaded files, with no intermediate manual editing of results.

Declarations

Availability of data and materials

TCGA-BRCA expression and clinical data are public and available from the UCSC Xena TCGA hub (`https://tcga.xenahubs.net`). No proprietary or restricted-access data were used in this paper.

Competing interests

The authors declare no competing interests.

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Appendix: full threshold sweep

Tables 6 and 7 give the complete sweep underlying Fig. 1 and Sec. , reproduced verbatim from `results/tnbc_rmt.csv` and `results/luma_rmt.csv`.

Table 6: Full TNBC threshold sweep.

τ	edges	density	KS(Poisson)	KS(GOE)	regime
0.15	383911	0.3415	0.1935	0.1050	GOE
0.20	238751	0.2124	0.2067	0.0949	GOE
0.25	142388	0.1267	0.2191	0.0772	GOE
0.30	82197	0.0731	0.2074	0.0947	GOE
0.35	46347	0.0412	0.2006	0.1272	GOE
0.40	25911	0.0230	0.1986	0.1670	GOE
0.45	**14498**	**0.0129**	**0.2504**	**0.2369**	**GOE** (τ^*)
0.50	7984	0.0071	0.3476	0.3919	Poisson
0.55	4391	0.0039	0.4246	0.5013	Poisson
0.60	2413	0.0021	0.4507	0.5428	Poisson
0.65	1389	0.0012	0.4715	0.5493	Poisson
0.70	814	0.0007	0.5442	0.6313	Poisson
0.75	478	0.0004	0.4712	0.5614	Poisson
0.80	258	0.0002	0.4217	0.4980	Poisson

Table 7: Full LumA threshold sweep.

τ	edges	density	KS(Poisson)	KS(GOE)	regime
0.15	347733	0.3093	0.2008	0.1106	GOE
0.20	223750	0.1990	0.1968	0.1222	GOE
0.25	**146100**	**0.1300**	**0.1899**	**0.1321**	**GOE** (τ^*)
0.30	98072	0.0872	0.1649	0.1653	Poisson
0.35	68007	0.0605	0.1763	0.1845	Poisson
0.40	47940	0.0426	0.1988	0.2305	Poisson
0.45	34153	0.0304	0.2587	0.2875	Poisson
0.50	24361	0.0217	0.3129	0.3706	Poisson
0.55	17034	0.0152	0.3902	0.4708	Poisson
0.60	11560	0.0103	0.4490	0.5313	Poisson
0.65	7427	0.0066	0.5182	0.6094	Poisson
0.70	4493	0.0040	0.5495	0.6333	Poisson
0.75	2553	0.0023	0.5911	0.6804	Poisson
0.80	1389	0.0012	0.4985	0.5881	Poisson